UniMentor — Unified AI-Powered Student Platform

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ABSTRACT

The design and development of an AI-powered unified student assistance platform, specifically focusing on personalized career guidance, exam preparation, skill development, and academic progress tracking. The primary objective of this project is to provide students with an intelligent and interactive learning ecosystem that enhances academic performance, career planning, and skill growth through adaptive AI technologies. The system operates using multi-provider AI services, where user interactions are processed through backend APIs, intelligent recommendation models, and analytics-driven workflows to generate personalized guidance, quizzes, learning paths, and progress insights. The document includes a study of existing educational and mentoring platforms, highlighting the need for a centralized AI-driven student support system. The methodology section outlines the systematic development process, including frontend and backend architecture design, AI integration, database management, user interaction handling, and analytics processing. Key design aspects such as Next.js frontend development, API routing, LangChain orchestration, AI service integration, MongoDB database management, and recommendation workflows are discussed in detail. This project aims to contribute to advancements in AI-powered education technology by providing a scalable, personalized, and engaging digital mentoring platform for students.

OUTCOMES

- Outcome 4
- Outcome 8
- Outcome 9

SDG ALIGNMENT

- SDG 4: Quality Education
- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure

Multi-Agent Real-Time Network Threat Detection and Automated Response System

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ABSTRACT

CyberAgent is an advanced Security Operations Center (SOC) automation platform designed to provide intelligent, real-time cybersecurity monitoring and response using a multi-agent architecture. Unlike traditional systems that rely on static rule-based detection or standalone machine learning approaches, CyberAgent integrates live application, authentication, and network telemetry with intelligent agent-based reasoning and pretrained threat detection models.

The system consists of a telemetry-generating client application, a backend collector and incident processing engine, a Multi-Component Platform (MCP) orchestration layer, and a real-time SOC dashboard. Incoming telemetry is normalized, analyzed, and correlated through specialized agents, while pretrained models enhance threat classification through feature mapping and inference.

CyberAgent employs a dynamic coordination mechanism that adaptively selects analysis and response stages, enabling flexible and efficient incident handling. A multi-agent review loop ensures accurate classification, validation, investigation, and policy-driven response. The system supports automated mitigation actions with human approval for critical scenarios, ensuring both efficiency and safety.

OUTCOMES

- Outcome 9
- Outcome 11
- Outcome 16

SDG ALIGNMENT

- SDG 9: Industry, Innovation and Infrastructure
- SDG 11: Sustainable Cities and Communication
- SDG 16: Peace, Justice and Strong Institutions

<table>
<tr><td></td><td>MINDMELD: AI AGENT BUILDER WEB APP</td><td></td></tr>
<tr><td>USN</td><td>STUDENTS NAME</td><td>GUIDE NAME</td></tr>
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</table>

ABSTRACT

Artificial intelligence is evolving from simple chat systems to agentic platforms capable of planning, executing tasks, and generating software solutions. MindMeld: AI Agent Builder Web App is a web-based platform that converts natural-language user requirements into structured AI-agent workflows and previewable software outputs. The system uses a master-agent architecture to decompose prompts into Directed Acyclic Graphs (DAGs) with task nodes, dependencies, and execution order. Workflow validation techniques such as dependency checking, cycle detection, and topological sorting ensure correctness before execution.

Built using Next.js, React, TypeScript, Tailwind CSS, AI SDK integrations, and sandbox environments, the platform supports code generation, workflow visualization, execution monitoring, generated file inspection, and application previews. Users can iteratively refine outputs through follow-up prompts, enabling rapid prototyping and AI-assisted software development. The project integrates artificial intelligence, workflow orchestration, and modern web technologies to simplify software creation and provide a foundation for future enhancements in testing, deployment, and collaborative development.

OUTCOMES

- Outcome 9
- Outcome 4
- Outcome 8

SDG ALIGNMENT

- SDG 9: Industry, Innovation and Infrastructure
- SDG 4: Quality Education
- SDG 8: Decent Work and Economic Growth

DESIGN AND DEVELOPMENT OF AN ANALOG ACTIVE NOISE CANCELLATION SYSTEM

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ABSTRACT

The rapid growth of the startup ecosystem has created an increasing demand for efficient, data-driven investment intelligence tools. Traditional financial analysis methods are time-consuming, heavily manual, and often inaccessible to individual investors and early-stage analysts. This project presents a Multi-Agent Startup Financial Analyser, an intelligent web-based platform that automates comprehensive startup evaluation using a coordinated system of specialized AI agents powered by Large Language Models (LLMs).

The system employs a MERN-inspired architecture — comprising a React (Vite) frontend, a Node.js/TypeScript backend, and a PostgreSQL database — integrated with the LangChain framework and OpenAI's GPT-4o-mini model. Upon entering a startup name, the platform orchestrates five autonomous agents in parallel: a Financial Agent that retrieves funding rounds, investor profiles, revenue estimates, and founder backgrounds; a SWOT Agent that performs strategic Strengths, Weaknesses, Opportunities, and Threats analysis; a Competitor Agent that maps the competitive landscape and benchmarks rivals; a Scoring Agent that applies venture-capital-standard weighted scoring across six investment dimensions (team strength, market size, traction, competitive moat, business model, and founder fit); and a Manager Agent that synthesizes all findings into a professional, citation-backed Markdown investment report.

OUTCOMES

- Outcome 8
- Outcome 9
- Outcome 10

SDG ALIGNMENT

- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure
- SDG 10: Reduced Inequalities